

SOC Internship Day 29

Supply Chain Security

Objective

Learn how SOC analysts detect, investigate, and respond to supply chain attacks.

What is Supply Chain Security?

Supply Chain Security focuses on protecting software, hardware, vendors, and third-party services from cyber threats. Attackers often target trusted suppliers to compromise multiple organizations at once.

Why Supply Chain Attacks Are Dangerous

- Target trusted software and services
- Can impact many organizations simultaneously
- Malware may spread through legitimate software updates
- Often bypass traditional security controls and defenses

Common Supply Chain Threats

1. Malicious Software Updates

Attackers inject malicious code into legitimate updates.

2. Compromised Vendors

A trusted vendor is breached and used to distribute malware.

3. Dependency Hijacking

Malicious code is inserted into software libraries or dependencies.

4. Fake Software Packages

Attackers publish fake packages that appear legitimate.

Indicators of a Supply Chain Attack

SOC analysts should look for:

- Suspicious activity after a software update
- Unexpected outbound connections
- Modified application files
- Unknown or unauthorized processes running on systems

Example Investigation

Alert

Suspicious activity detected after a software update.

Findings

- Application connected to a malicious domain
- Unknown DLL files were found on the system

Analysis

Possible supply chain compromise through a malicious update.

Detection Techniques

- Verify digital signatures of software
- Monitor software integrity
- Analyze update behavior
- Track vendor activity and reputation

SOC Workflow for Supply Chain Incidents

1. Software Update Installed
2. Suspicious Activity Detected
3. SOC Investigation
4. Containment and Response

SOC Team Roles

L1 Analyst

- Monitor security alerts
- Identify suspicious activities

L2 Analyst

- Investigate affected applications
- Validate indicators of compromise

L3 Analyst

- Perform threat hunting
- Lead incident response and remediation

Key Takeaways

- Supply chain attacks exploit trusted software, vendors, or services.
- They can affect thousands of organizations through a single compromise.
- Monitoring software updates and verifying digital signatures are critical defenses.

- SOC teams must quickly investigate suspicious behavior following updates.
- Early detection and containment help minimize damage.

Assignment Answers

1. Explain Supply Chain Security

Supply Chain Security is the practice of protecting software, hardware, vendors, and third-party services from cyber threats. Its goal is to prevent attackers from compromising trusted components and distributing malicious content to organizations.

2. Research Famous Supply Chain Attacks

- SolarWinds Supply Chain Attack
- Kaseya VSA Attack
- CCleaner Supply Chain Attack

3. Analyze a Malicious Update Scenario

A software vendor unknowingly releases an update containing malicious code. After installation, systems begin communicating with suspicious external domains and unauthorized DLL files appear. SOC analysts investigate the activity, confirm the compromise, isolate affected systems, and remove the malicious software to prevent further spread.